YouTube Data Pipeline (ETL) & Analytics Dashboard

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1. Introduction

With the exponential growth of video content platforms, analyzing trends on YouTube has become essential for understanding user behavior and content performance. This project builds a complete **data pipeline** to extract, process, and visualize trending YouTube video data.

2. Objective

The objectives of this project are:

- Extract trending video data using YouTube API
- Transform raw data into a clean and structured format
- Store data in a relational database (MySQL)
- Perform SQL-based analysis
- Build an interactive Power BI dashboard
- Generate meaningful insights for decision-making

3. Problem Statement

YouTube generates massive amounts of unstructured data. Manually analyzing trending videos is inefficient and time-consuming.

This project aims to:

- Automate data collection
- Clean and structure raw data
- Identify trends in views, likes, and engagement
- Provide interactive visual insights

4. Tools & Technologies

- **Python** (Pandas, API integration)
- **YouTube Data API v3**

- **MySQL** (Database)
- **SQL** (Data analysis)
- **Power BI** (Dashboard & visualization)

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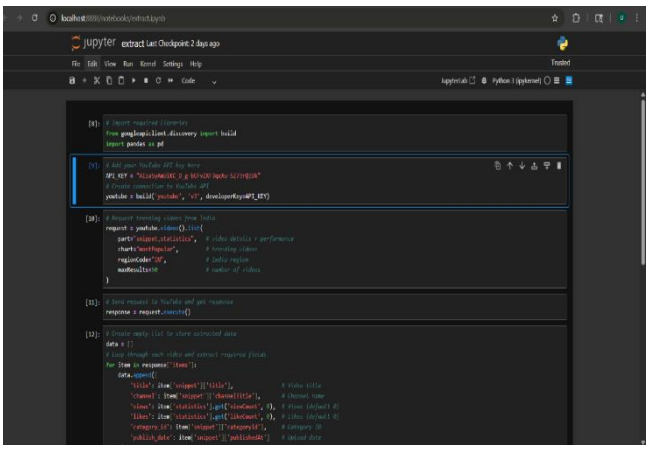
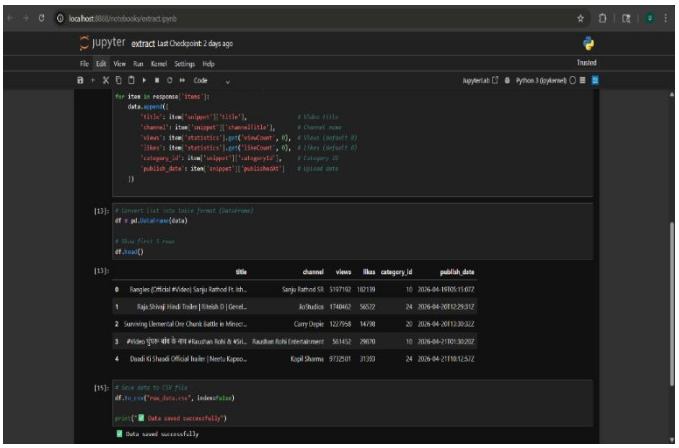
5. System Architecture (ETL Pipeline)

This project follows the **ETL (Extract, Transform, Load)** process:

5.1 Extract

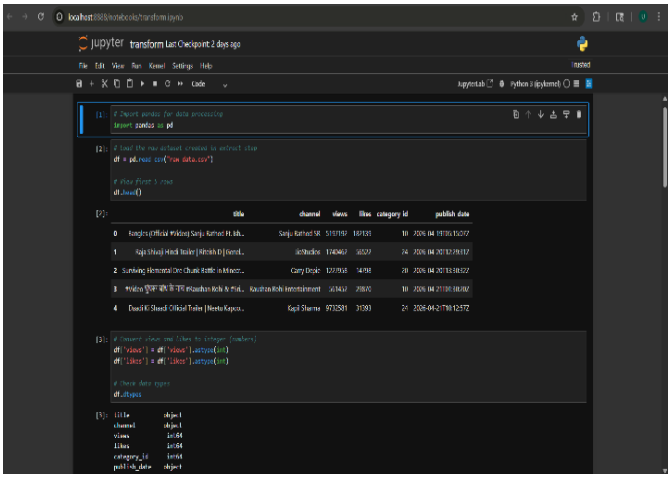
- Data fetched using YouTube API
- Trending videos from India region
- Fields collected:
    - Title
    - Channel
    - Views
    - Likes
    - Category ID
    - Publish Date

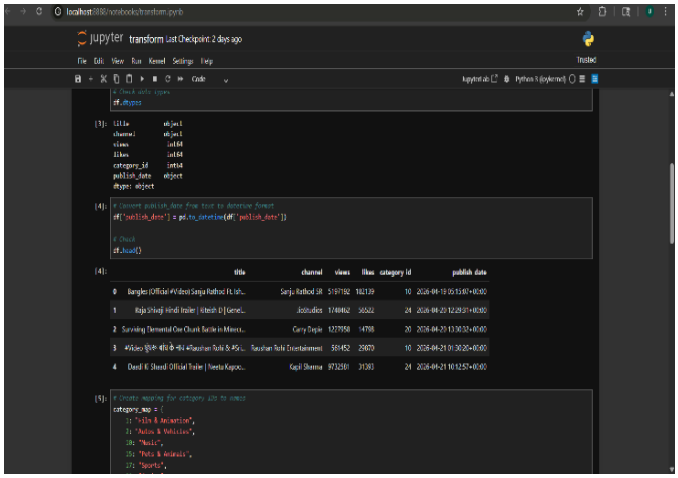


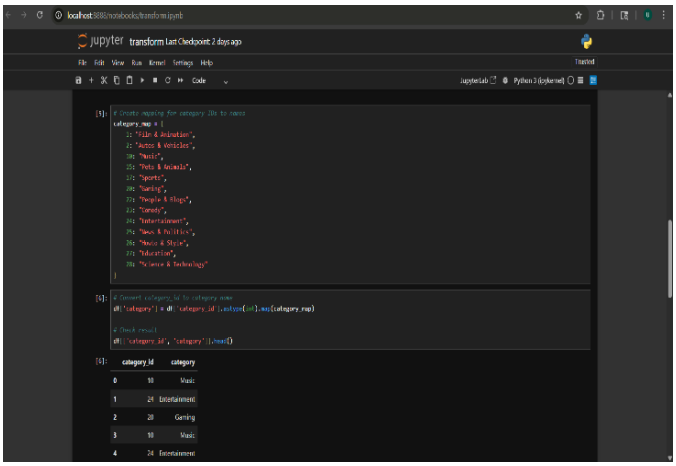


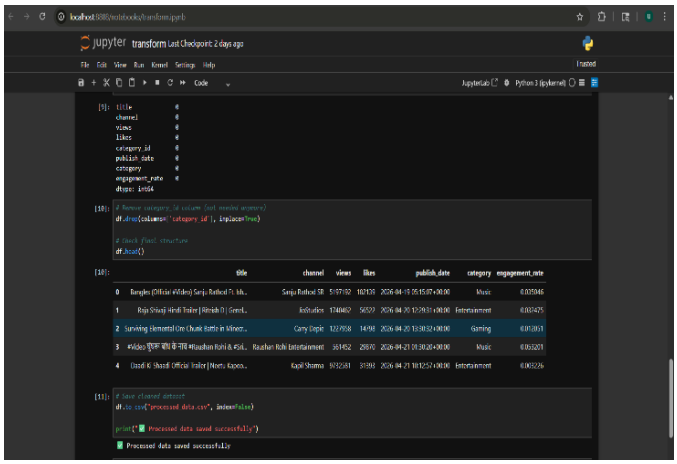
5.2 Transform

- Data cleaning using Pandas
- Converted views & likes to numeric
- Converted publish_date to datetime
- Mapped category_id to category names
- Created new column: **engagement_rate = likes / views**
- Removed null values



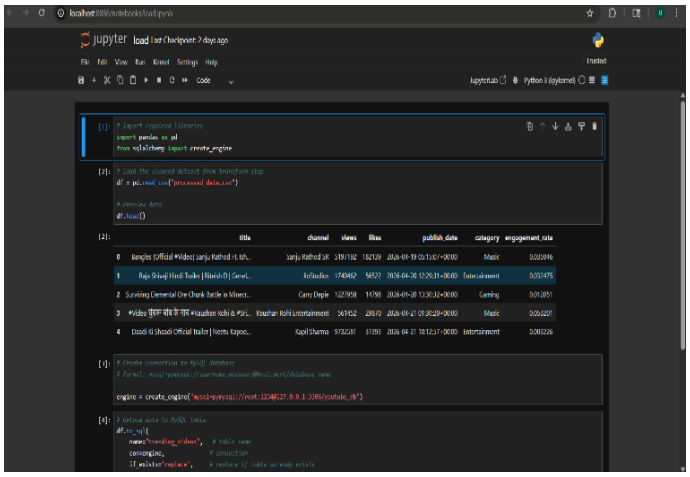


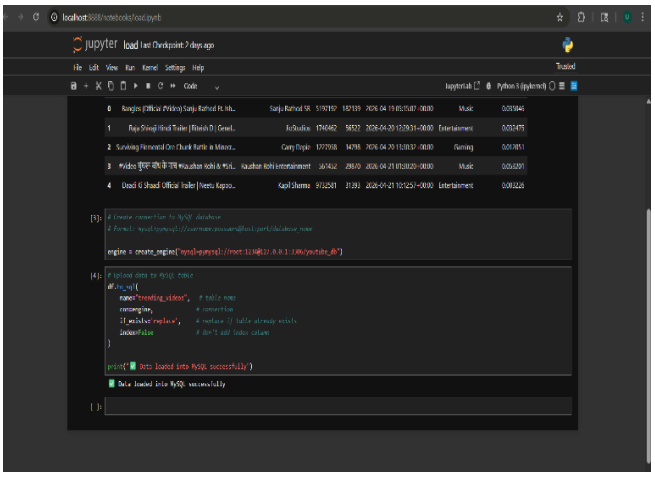




5.3 Load

- Cleaned data stored in MySQL
- Table name: trending_videos
- Enabled efficient querying and analysis



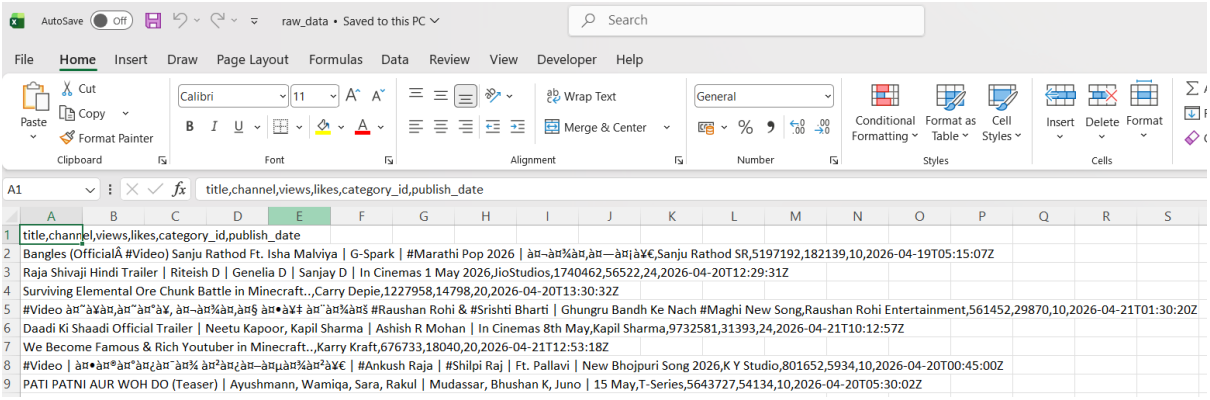


6. Dataset Description

6.1 Raw Dataset

File: raw_data.csv

- Direct output from API
- Unstructured and inconsistent
- Contains category IDs and text dates



```
title,channel,views,likes,category_id,publish_date
Bangles (OfficialÂ #Video) Sanju Rathod Ft. Isha Malviya | G-Spark | #Marathi Pop 2026 | à¤¬à¤¾à¤à¤¦à¤¿à¥€,Sanju Rathod SR,5197192,182139,10,2026-04-19T05:15:07Z
Raja Shivaji Hindi Trailer | Riteish D | Genelia D | Sanjay D | In Cinemas 1 May 2026,JioStudios,1740462,56522,24,2026-04-20T12:29:31Z
Surviving Elemental Ore Chunk Battle in Minecraft..,Carry Depie,1227958,14798,20,2026-04-20T13:30:32Z
#Video à¤˜à¥à¤,à¤¯à¤°à¥,à¤¬à¤¾à¤§ à¤•à¥‡ à¤¨à¤¼à¤à¤š #Raushan Rohi & #Srishti Bharti | Ghungru Bandh Ke Nach #Maghi New Song,Raushan Rohi Entertainment,561452,29870,10,2026-04-21T01:30:20Z
Daadi Ki Shaadi Official Trailer | Neetu Kapoor, Kapil Sharma | Ashish R Mohan | In Cinemas 8th May,Kapil Sharma,9732581,31393,24,2026-04-21T10:12:57Z
We Become Famous & Rich Youtuber in Minecraft..,Karry Kraft,676733,18040,20,2026-04-21T12:53:18Z
#Video | à¤•à¤®à¤°à¤¿à¤¯à¤¾à¤² à¤²à¤¿à¤–à¤µà¤¾à¤²à¥€ | #Ankush Raja | #Shilpi Raj | Ft. Pallavi | New Bhojpuri Song 2026,K Y Studio,801652,5934,10,2026-04-20T00:45:00Z
PATI PATNI AUR WOH DO (Teaser) | Ayushmann, Wamiqa, Sara, Rakul | Mudassar, Bhushan K, Juno | 15 May,T-Series,5643727,54134,10,2026-04-20T05:30:02Z
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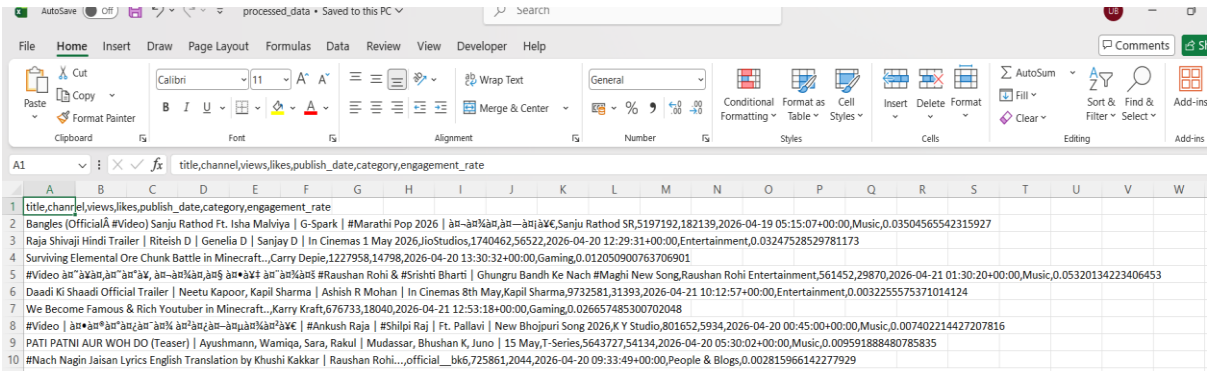
6.2 Processed Dataset

File: processed_data.csv

- Cleaned and structured
- Ready for analysis
- Includes new feature: engagement_rate



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title,channel,views,likes,publish_date,category,engagement_rate
Bangles (OfficialÂ #Video) Sanju Rathod Ft. Isha Malviya | G-Spark | #Marathi Pop 2026 | à¤¬à¤¾à¤à¤¦à¤¿à¥€,Sanju Rathod SR,5197192,182139,2026-04-19 05:15:07+00:00,Music,0.03504565542315927
Raja Shivaji Hindi Trailer | Riteish D | Genelia D | Sanjay D | In Cinemas 1 May 2026,JioStudios,1740462,56522,2026-04-20 12:29:31+00:00,Entertainment,0.03247528529781173
Surviving Elemental Ore Chunk Battle in Minecraft..,Carry Depie,1227958,14798,2026-04-20 13:30:32+00:00,Gaming,0.012050900763706901
#Video à¤˜à¥à¤,à¤¯à¤°à¥,à¤¬à¤¾à¤§ à¤•à¥‡ à¤¨à¤¼à¤à¤š #Raushan Rohi & #Srishti Bharti | Ghungru Bandh Ke Nach #Maghi New Song,Raushan Rohi Entertainment,561452,29870,2026-04-21 01:30:20+00:00,Music,0.05320134223406453
Daadi Ki Shaadi Official Trailer | Neetu Kapoor, Kapil Sharma | Ashish R Mohan | In Cinemas 8th May,Kapil Sharma,9732581,31393,2026-04-21 10:12:57+00:00,Entertainment,0.003225557537101412 4
We Become Famous & Rich Youtuber in Minecraft..,Karry Kraft,676733,18040,2026-04-21 12:53:18+00:00,Gaming,0.026657485300702048
#Video | à¤•à¤®à¤°à¤¿à¤¯à¤¾à¤² à¤²à¤¿à¤–à¤µà¤¾à¤²à¥€ | #Ankush Raja | #Shilpi Raj | Ft. Pallavi | New Bhojpuri Song 2026,K Y Studio,801652,5934,2026-04-20 00:45:00+00:00,Music,0.007402214427207816
PATI PATNI AUR WOH DO (Teaser) | Ayushmann, Wamiqa, Sara, Rakul | Mudassar, Bhushan K, Juno | 15 May,T-Series,5643727,54134,2026-04-20 05:30:02+00:00,Music,0.009591888480785835
#Nach Nagin Jaisan Lyrics English Translation by Khushi Kakkar | Raushan Rohi...,official__bk6,725861,2044,2026-04-20 09:33:49+00:00,People & Blogs,0.002815966142277929
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6.3 Difference

Feature	Raw Data	Processed Data
Format	Unstructured	Structured
Category	ID	Name

Data Type	Mixed	Clean
Missing Values	Present	Removed
Analysis Ready	No	Yes

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7. Database Design

Table Name: trending_videos

Column	Description
title	Video title
channel	Channel name
views	Number of views
likes	Number of likes
category	Video category
publish_date	Upload date
engagement_rate	Likes / Views

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

title	channel	views	likes	publish_date	category	engagement_rate
Bangles (Official #Video) Sanju Rathod Ft. Isha ...	Sanju Rathod SR	5197192	182139	2026-04-19 05:15:07+00:00	Music	0.0350456554231592
Raja Shivaji Hindi Trailer | Riteish D | Genelia D |...	JioStudios	1740462	56522	2026-04-20 12:29:31+00:00	Entertainment	0.0324752852978117
Surviving Elemental Ore Chunk Battle in Minecra...	Carry Depie	1227958	14798	2026-04-20 13:30:32+00:00	Gaming	0.0120509007637069
#Video घुंघरू बांध के नाच #Raushan Rohi & #Sris...	Raushan Rohi Entertainment	561452	29870	2026-04-21 01:30:20+00:00	Music	0.0532013422340645
Daadi Ki Shaadi Official Trailer | Neetu Kapoor, ...	Kapil Sharma	9732581	31393	2026-04-21 10:12:57+00:00	Entertainment	0.0032255575371014
We Become Famous & Rich Youtuber in Minecra...	Karry Kraft	676733	18040	2026-04-21 12:53:18+00:00	Gaming	0.026657485300702
#Video | कमरिया लिखवाली | #Ankush Raja | #Sh...	K Y Studio	801652	5934	2026-04-20 00:45:00+00:00	Music	0.0074022144272078

ing_videos1 x

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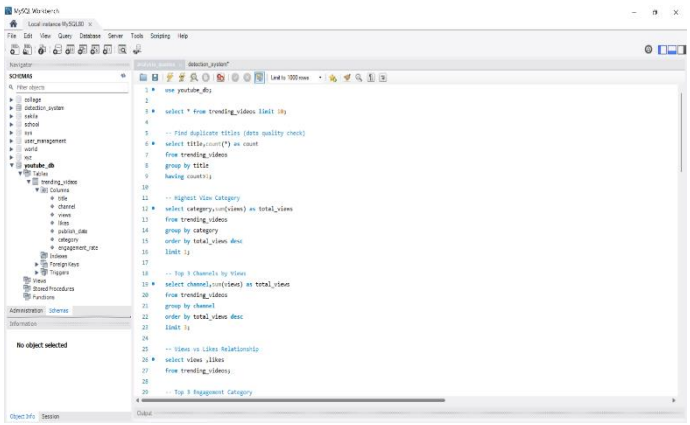
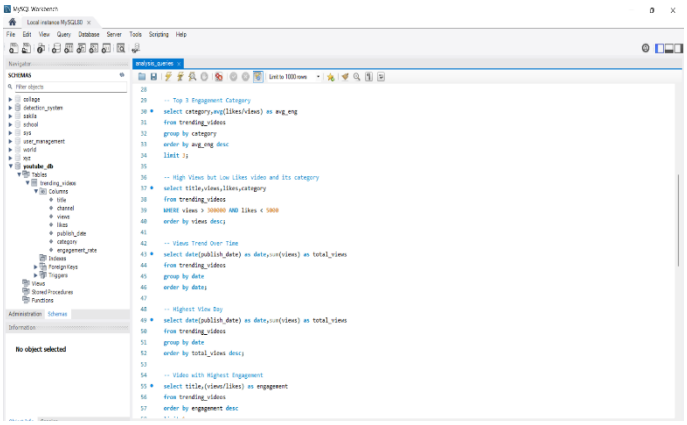
8. SQL Analysis

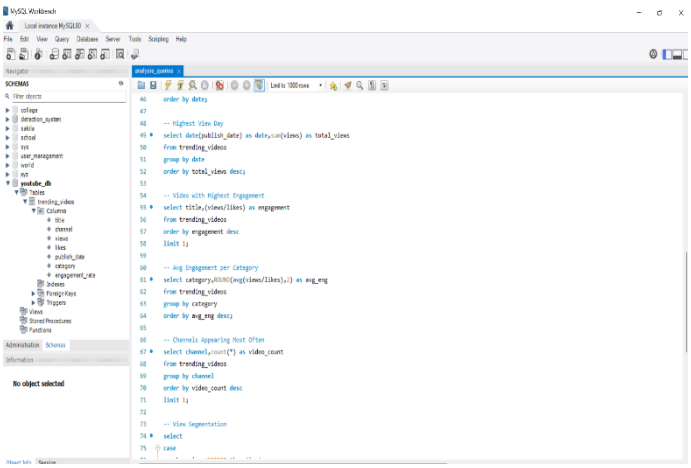
Key queries performed:

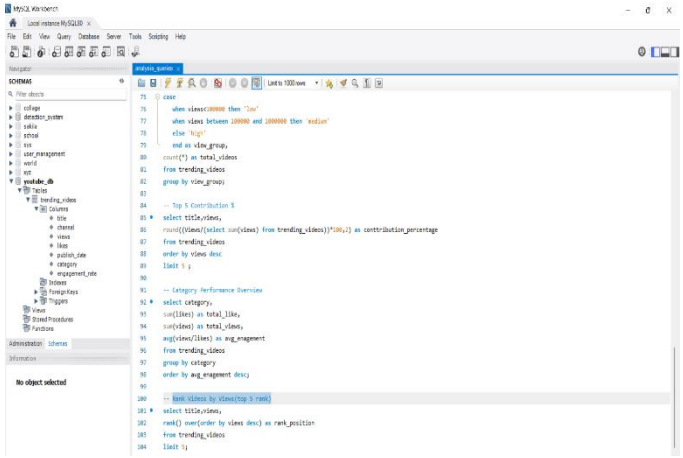
- Find duplicate titles (data quality check)
- Highest View Category
- Top 3 Channels by Views
- Views vs Likes Relationship
- Top 3 Engagement Category
- High Views but Low Likes video and its category

- Views Trend Over Time
- Highest View Day
- Video with Highest Engagement
- Avg Engagement per Category
- Channels Appearing Most Often
- View Segmentation
- Top 5 Contribution %
- Category Performance Overview
- Rank Videos by Views (top 5 rank)









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9. Power BI Dashboard

9.1 Pages:

- **Home Page** → Overview
- **Details Page** → Table-level analysis

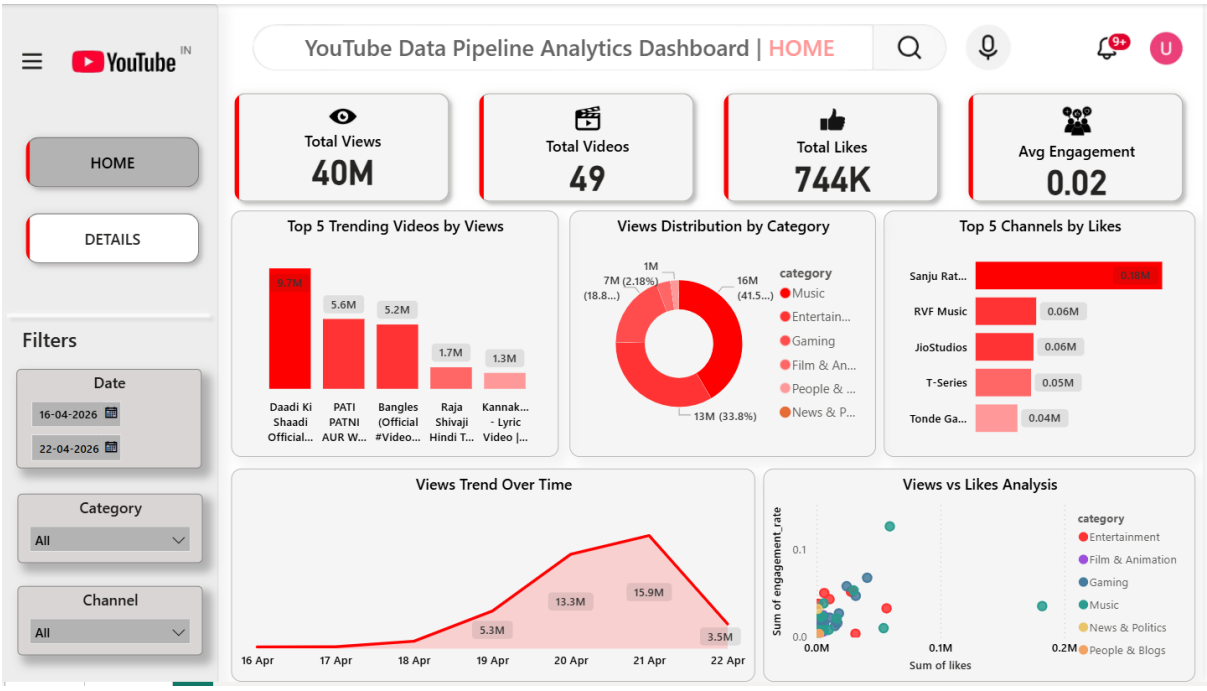
9.2 Visualizations Used:

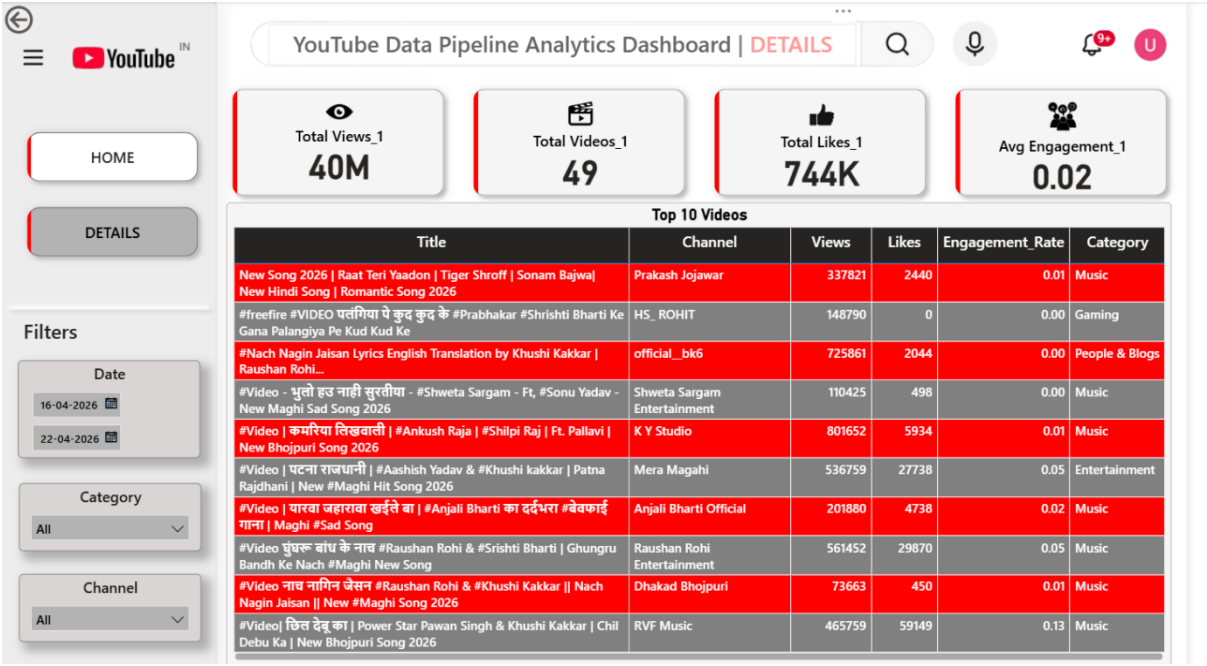
- KPI Cards:
  - Total Views
  - Total Likes
  - Total Videos
  - Avg Engagement Rate
- Charts:
  - Top Videos by Views (Bar Chart)
  - Category Distribution (Donut Chart)
  - Top Channels by Likes (Bar Chart)
  - Views Trend Over Time (Line Chart)
  - Views vs Likes (Scatter Plot)

9.3 Filters:

- Date
- Category
- Channel

9.4 Dashboard Preview





Title	Channel	Views	Likes	Engagement_Rate	Category
New Song 2026 | Raat Teri Yaadon | Tiger Shroff | Sonam Bajwa| New Hindi Song | Romantic Song 2026	Prakash Jojawar	337821	2440	0.01	Music
#freefire #VIDEO पतंगिया पे कुद कुद के #Prabhakar #Shrishti Bharti Ke Gana Palangiya Pe Kud Kud Ke	HS_ ROHIT	148790	0	0.00	Gaming
#Nach Nagin Jaisan Lyrics English Translation by Khushi Kakkar | Raushan Rohi...	official_bk6	725861	2044	0.00	People & Blogs
#Video - भुलो हउ नाही सुरतीया - #Shweta Sargam - Ft, #Sonu Yadav - New Maghi Sad Song 2026	Shweta Sargam Entertainment	110425	498	0.00	Music
#Video | कमरिया तिखवाती | #Ankush Raja | #Shilpi Raj | Ft. Pallavi | New Bhojpuri Song 2026	K Y Studio	801652	5934	0.01	Music
#Video | पटना राजधानी | #Aashish Yadav & #Khushi kakkar | Patna Rajdhani | New #Maghi Hit Song 2026	Mera Magahi	536759	27738	0.05	Entertainment
#Video | यारवा जहारावा खईते बा | #Anjali Bharti का दर्दभरा #बेवफाई गाना | Maghi #Sad Song	Anjali Bharti Official	201880	4738	0.02	Music
#Video घुंघरू बांध के नाच #Raushan Rohi & #Srishti Bharti | Ghungru Bandh Ke Nach #Maghi New Song	Raushan Rohi Entertainment	561452	29870	0.05	Music
#Video नाच नागिन जेसन #Raushan Rohi & #Khushi Kakkar || Nach Nagin Jaisan || New #Maghi Song 2026	Dhakad Bhojpuri	73663	450	0.01	Music
#Video| छित देबू का | Power Star Pawan Singh & Khushi Kakkar | Chil Debu Ka | New Bhojpuri Song 2026	RVF Music	465759	59149	0.13	Music

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10. Key Insights

- A small number of videos generate the majority of views
- Music and Entertainment dominate trending content
- High views do not always mean high engagement
- Some low-view videos have higher engagement rates
- Channel performance varies significantly

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11. Challenges Faced

- Handling missing API data
- Converting incorrect data types
- Fixing date format issues
- Managing database connectivity
- Designing visually clean dashboard

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12. Future Enhancements

- Automate pipeline using scheduler
- Store historical data for trend analysis
- Add real-time data updates

- Implement machine learning for prediction
- Deploy dashboard online

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13. Conclusion

This project demonstrates a complete data analytics workflow from data extraction to visualization. It highlights the importance of data cleaning, structured storage, and interactive dashboards in deriving meaningful insights.

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15. Project Structure

```
youtube-etl-project/
│
├── data/
│   ├── raw_data.csv
│   ├── processed_data.csv
│
├── scripts/
│   ├── extract.py
│   ├── transform.py
│   ├── load.py
│
├── sql_queries.sql
├── dashboard.pbix
├── README.md
```

Project link :- https://github.com/urmil24/youtube-etl-project

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